

Coarse-Grained Entropy and Exterior Records: Compatible Microstates and a Reconstruction Boundary for Semiclassical Black Holes

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Let $\pi : \Gamma \rightarrow M$ be a declared coarse-graining from microscopic states to macrostates. For a macrostate m , the compatible-state fiber $\mathcal{E}(m) = \pi^{-1}(m)$ retains the microscopic distinctions unresolved by that read, and a supplied measure μ gives the Boltzmann multiplicity $\Omega(m) = \mu(\mathcal{E}(m))$ and entropy $S_B(m) = k_B \ln \Omega(m)$. We isolate the corresponding reconstruction boundary: if an upstream record map identifies two histories, every downstream reader of that record identifies them as well. Applied to a black-hole exterior record, this gives a limited statement about record resolution only. Hawking temperature and Bekenstein-Hawking entropy are taken as semiclassical inputs; no microscopic state count, evaporation model, or quantum-gravity dynamics is derived. We close by locating the result relative to the loop-quantum-gravity black-hole entropy program, whose quantum-geometric microstate constructions address a strictly deeper input problem than the record theorem considered here.

I. COARSE-GRAINING AND COMPATIBLE MICROSTATES

Let Γ be a microscopic state space and M a macroscopic state space. A coarse-graining is a declared map

$$\pi : \Gamma \longrightarrow M. \quad (1)$$

For $\gamma \in \Gamma$, write $m = \pi(\gamma)$. The states compatible with the same macro-read form the fiber

$$\boxed{\mathcal{E}(m) := \pi^{-1}(m).} \quad (2)$$

Thus $\pi(\gamma_1) = \pi(\gamma_2)$ places γ_1 and γ_2 in one macrostate class. It leaves their microscopic equality as a separate question.

Let μ be the physical counting measure, phase-space measure, or other measure appropriate to the admitted statistical model. When

$$0 < \Omega(m) := \mu(\mathcal{E}(m)) < \infty, \quad (3)$$

the Boltzmann entropy is

$$\boxed{S_B(m) = k_B \ln \Omega(m).} \quad (4)$$

The logical order is therefore

$$\begin{aligned} \text{microstate carrier} &\xrightarrow{\pi} \text{macrostate} \longrightarrow \text{compatible fiber} \\ &\longrightarrow \text{multiplicity} \longrightarrow \text{entropy}. \end{aligned} \quad (5)$$

Equation (4) is a numerical read of the multiplicity left compatible with the declared macrostate. The map π , the fiber $\mathcal{E}(m)$, the measure μ , and the number $S_B(m)$ occupy distinct roles.

a. Proposition 1 (macrostate resolution). If $\pi(\gamma_1) = \pi(\gamma_2)$, every quantity that factors only through π assigns the same value to γ_1 and γ_2 .

Proof. For any $r : M \rightarrow Y$,

$$r(\pi(\gamma_1)) = r(\pi(\gamma_2)). \quad (6)$$

The statement is simply functoriality of equality through the downstream map r . $\square$

This elementary point becomes load-bearing whenever a coarse description is promoted into a claim about the microscopic history that produced it.

II. THE RECORD-RECONSTRUCTION BOUNDARY

Let $\mathcal{X}$ be a class of states or histories, let Σ be a record space, and let

$$P : \mathcal{X} \longrightarrow \Sigma \quad (7)$$

be the map that writes the record available to a downstream observer or procedure.

a. Theorem 2 (record no-reconstruction). Suppose $x_1, x_2 \in \mathcal{X}$ satisfy

$$P(x_1) = P(x_2). \quad (8)$$

Then every downstream reader $R : \Sigma \rightarrow Y$ satisfies

$$\boxed{R(P(x_1)) = R(P(x_2)).} \quad (9)$$

Consequently, if a promised target read $\tau : \mathcal{X} \rightarrow Z$ distinguishes the pair,

$$\tau(x_1) \neq \tau(x_2), \quad (10)$$

there is no map $R : \Sigma \rightarrow Z$ such that $R \circ P = \tau$ on both x_1 and x_2 .

Proof. Equation (8) and functionality of R give Eq. (9). If $R \circ P = \tau$ held on both inputs, Eq. (9) would imply $\tau(x_1) = \tau(x_2)$, contradicting the stated target distinction. $\square$

The theorem concerns the resolving power of a specified record. It says nothing about distinctions retained elsewhere in the full dynamics, nor about whether a richer record map exists.

III. SEMICLASSICAL BLACK-HOLE SPECIALIZATION

For an asymptotically appropriate spacetime, write the black-hole region and future event horizon as

$$\mathcal{B}_{\text{BH}} = M_{\text{spacetime}} \setminus J^-(\mathcal{I}^+), \quad \mathcal{H}_{\text{BH}}^+ = \partial J^-(\mathcal{I}^+). \quad (11)$$

Classical black-hole mechanics supplies the horizon first-law structure [1]; quantum field theory on the black-hole background supplies the Hawking temperature [3], and the semiclassical entropy law reads [2, 3]

$$T_H = \frac{\kappa_H}{2\pi}, \quad \boxed{S_{\text{BH}} = \frac{A_H}{4}} \quad (12)$$

in natural units $G = \hbar = c = k_B = 1$.

Equation (12) is used here as an input relation. It assigns an entropy to the semiclassical horizon area. Complete recovery of an interior history is a different problem because it requires a record map and a decoder with the corresponding resolution.

Let

$$P_{\text{ext}} : \mathcal{X}_{\text{full}} \longrightarrow \Sigma_{\text{ext}} \quad (13)$$

be any explicitly specified exterior-record map. If

$$P_{\text{ext}}(h_1) = P_{\text{ext}}(h_2), \quad (14)$$

then Theorem 2 gives, for every reader $R_{\text{ext}} : \Sigma_{\text{ext}} \rightarrow Y$,

$$R_{\text{ext}} P_{\text{ext}}(h_1) = R_{\text{ext}} P_{\text{ext}}(h_2). \quad (15)$$

Thus the specified exterior record cannot by itself reconstruct a distinction already identified by its own upstream map.

This conclusion is deliberately narrower than a statement about fundamental information loss. Unitary evolution, tracing over inaccessible degrees of freedom, evaporation, Page-curve behavior, and microscopic horizon dynamics are different physical questions. The theorem fixes only the information available from the declared record P_{ext} .

IV. MICROSCOPIC BOUNDARY: LOOP QUANTUM GRAVITY

Loop quantum gravity addresses an upstream problem that the preceding argument leaves supplied: what quantum-geometric microscopic degrees of freedom account for black-hole entropy? Early LQG state-counting constructions identify horizon degrees of freedom and recover an entropy proportional to area [4, 5]; subsequent work sharpened the counting of horizon states [6]. Han’s analytic-continuation construction instead defines a black-hole partition function in LQG and recovers the Bekenstein-Hawking leading term from the quantum-geometric framework [7].

The relation between the two layers can therefore be stated positively:

$$\boxed{\text{QG microstates} \longrightarrow \text{state count} \longrightarrow S_{\text{BH}}}, \quad (16)$$

$$\boxed{\text{exterior record} \longrightarrow \text{record-limited reconstruction}}. \quad (17)$$

The first line is the microscopic construction problem. The second is the record-resolution theorem proved here. A complete black-hole information theory requires additional dynamical structure connecting these layers; no such completion is asserted in this note.

V. CONCLUSION

A coarse-graining map determines a fiber of compatible microscopic states. Boltzmann entropy reads the measured multiplicity of that fiber. A record map determines which distinctions remain available to every downstream reader of that record. At a semiclassical black-hole horizon, these facts supply a precise bookkeeping boundary while leaving the microscopic origin of S_{BH} and the dynamics of quantum information to the quantum-gravity theory that carries them.

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